Metal artifact reduction (MAR) methods are increasingly incorporated into computed tomography and other tomographic imaging systems. These methods may either improve or degrade lesion detectability and quantitative accuracy depending on imaging parameters and algorithm design. Current standards do not define an objective, task-based, reproducible, interlaboratory test method for quantifying this effect.

This test method establishes a procedure for measuring MAR performance using a standardized digital volumetric dataset, a single canonical metal and lesion configuration, and a channelized Hotelling observer (CHO) model. The figure of merit is ΔAUC: the signed difference in area under the ROC curve between MAR-enabled and MAR-disabled conditions. The digital, checksum-verified dataset eliminates physical phantom logistics, permitting reproducible interlaboratory comparison. The method applies to systems producing HU-calibrated reconstructed image data and is structured to support precision and bias evaluation in accordance with ISO 5725. It is intended for type testing and does not establish performance acceptance criteria.

Standard Test Method for Quantitative Evaluation of Metal Artifact Reduction Performance in Tomographic Imaging Systems[1]

This standard is issued under the fixed designation X XXXX; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

1. Scope

[1] This test method is under the jurisdiction of ASTM Committee F04 on Medical and Surgical Materials and Devices and is the direct responsibility of Subcommittee TBD.

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1.1 This test method specifies a quantitative procedure for evaluating the performance of Metal Artifact Reduction (MAR) methods implemented in tomographic imaging systems.

1.2 The method applies to tomographic imaging systems that produce reconstructed volumetric image data.

1.3 The measurand is the scalar difference in area under the receiver operating characteristic curve (ΔAUC) derived from a signal-known-exactly, background-known-statistically (SKE/BKS) detection task using a fixed channelized Hotelling observer (CHO) implementation, predefined regions of interest, and specified statistical estimation procedures as defined in Annex A1.

1.4 This method is modality-independent in the sense that the measurand (ΔAUC) is computed exclusively from reconstructed volumetric image data and does not depend on acquisition hardware, raw projection data, or reconstruction physics. The canonical dataset (§10.1.1) is defined in Hounsfield Unit (HU) space and is therefore directly applicable to systems that produce HU-calibrated reconstructed images, including but not limited to computed tomography (CT). Application to modalities that do not produce HU-calibrated output requires a modality-specific canonical dataset established by a separate work item; results from such datasets shall not be reported as compliant with this standard unless the relevant modality annex has been approved.

1.5 This standard defines a single canonical test configuration intended for type testing and conformity assessment. No additional configurations are permitted under this standard.

1.6 *Units*—The values stated in SI units are to be regarded as the standard. No other units of measurement are included in this standard.

1.7 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.8 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

1A.1 Background and Technical Basis (Informational)

1A.1 Metal artifacts in CT arise primarily from beam hardening, photon starvation, and partial volume effects in the presence of highly attenuating objects. MAR algorithms mitigate these artifacts but may alter image statistics in ways that affect clinically relevant tasks, including low-contrast lesion detection. The net effect of MAR on diagnostic utility is not captured by artifact severity metrics alone.

1A.2 The framework underlying this standard was first described in *Vaishnav, et al.* (Medical Physics, 47(8), 2020), which demonstrated that a model observer-based approach could objectively measure MAR effects on low-contrast detectability. The channelized Hotelling observer (CHO) is a linear model observer that has been extensively validated against human performance in CT detection tasks. The area under the ROC curve (AUC), estimated via the Mann-Whitney statistic, provides a scalar figure of merit that is interpretable, reproducible, and independent of decision threshold. Building upon this framework, the reference baseline AUC_noMAR = 0.7063 (LOO hold-out; 95% CI [0.6575, 0.7844]) was established by utilizing sinogram-domain physics contrast (~12 HU). This validated result confirms that the task is noise-limited and metrologically sensitive to Metal Artifact Reduction quality, consistent with human-correlated task-based assessment paradigms.

1A.3 The use of a deterministic digital dataset eliminates physical phantom fabrication, scanner time, and shipping logistics from interlaboratory studies. SHA-256 checksums ensure bitwise identity of the dataset across laboratories. The only source of interlaboratory variability under this standard is therefore the CHO implementation and the MAR algorithm under test, which is the intended behavior.

1A.4 The canonical dataset provides background variability through per-realization rotation of the artifact template (§A1.7). This mechanism prevents the CHO from exploiting static artifact patterns and ensures that the observer's performance reflects genuine signal detectability rather than artifact fingerprint memorization.

2. Referenced Documents

2.1 *ASTM Standards:*

- *ASTM E177 – Practice for Use of the Terms Precision and Bias in ASTM Test Methods*
- *ASTM E691 – Practice for Conducting an Interlaboratory Study to Determine the Precision of a Test Method.*

2.2 *ISO Standards:*

- *ISO 5725-1 – Accuracy (trueness and precision) of measurement methods and results – Part 1.*
- *ISO 5725-2 – Basic method for determination of repeatability and reproducibility of a standard measurement method*

2.3 *Other References:*

- *Vaishnav JY et al. CT metal artifact reduction algorithms: Toward a framework for objective performance assessment. Medical Physics 47(8):3344–3355, 2020. DOI: 10.1002/mp.14231*

- *Barrett HH, Myers KJ. Foundations of Image Science. Wiley, 2004.*
- *FIPS PUB 180-4 — Secure Hash Standard (SHA-256)*

3. Terminology

3.1 *Definitions:*

3.1.1 metal artifact—image distortion in reconstructed tomographic data arising from the presence of a high-attenuation object, manifesting as streak, shadow, or halo patterns.

3.1.2 metal artifact reduction (MAR)—algorithmic processing applied within the imaging system pipeline to mitigate metal artifacts in reconstructed image data.

3.1.3 channelized Hotelling observer (CHO)—a linear model observer that evaluates signal detectability by projecting image data onto predefined spatial-frequency channel templates and applying the Hotelling discriminant to the resulting channel output vectors.

3.1.4 area under the ROC curve (AUC)—a scalar measure of binary detection performance ranging from 0.5 (chance-level discrimination) to 1.0 (perfect discrimination), estimated in this standard via the Mann–Whitney statistic.

3.1.5 ΔAUC—the signed difference $AUC_{MAR} - AUC_{noMAR}$, where positive values indicate that MAR improved lesion detectability and negative values indicate degradation.

3.1.6 test result—the mean ΔAUC calculated across all replicate paired image sets under specified conditions, reported to three decimal places.

3.1.7 signal-known-exactly (SKE)—a detection task paradigm in which the signal (lesion) location, shape, size, and intensity are fixed and known to the observer.

3.1.8 background-known-statistically (BKS)—a detection task paradigm in which the background statistics are known but individual background realizations vary. In this standard, variation arises from per-realization artifact jitter and independent Gaussian noise.

3.1.9 canonical test configuration—the single, fully specified combination of phantom geometry, lesion parameters, dataset generation rules, and observer implementation defined in this standard. No other configuration is permitted.

3.1.10 realization—one independent volumetric image set (lesion-present or lesion-absent) generated with a unique random seed, constituting an independent sample for CHO training and testing.

3.1.11 artifact jitter—per-realization rotation of the artifact template by a specified random angle, providing background variability that prevents CHO overtraining on static artifact patterns.

4. Summary of Test Method

4.1 A standardized reconstructed digital volumetric dataset is provided to participating laboratories. The dataset represents a simplified anthropomorphic torso cross-section containing a single cylindrical metallic rod and background tissue. The dataset is fully synthetic, generated from defined geometric primitives, and does not require clinical scanner acquisition.

4.2 The standardized volumetric dataset shall be processed: (a) with MAR disabled, and (b) with MAR enabled, using identical reconstruction, preprocessing, postprocessing, and display parameters. The MAR enable/disable state shall be the only permitted difference between conditions.

4.3 A specified cylindrical lesion rod is present in a defined spatial relationship to the metallic object in the lesion-present series. The lesion-absent series contains no lesion.

4.4 A two-dimensional channelized Hotelling observer (2D CHO) analysis is performed on paired lesion-present and lesion-absent image volumes for each condition (MAR and noMAR).

4.5 AUC values are computed for each condition via the Mann–Whitney statistic.

4.6 $\Delta AUC = AUC_{MAR} - AUC_{noMAR}$.

4.7 The mean ΔAUC across replicate image sets constitutes the test result.

5. Significance and Use

5.1 This test method provides an objective and reproducible means of quantifying the impact of MAR algorithms on diagnostic task performance preservation.

5.2 The method supports comparison of MAR implementations across systems and laboratories by measuring the preservation of lesion detectability in a standardized SKE/BKS detection task.

5.3 The test method is intended for type testing, research, and performance characterization.

5.4 This test method does not establish performance acceptance criteria. A positive ΔAUC indicates that MAR improves lesion detectability relative to the no-MAR condition; a negative ΔAUC indicates degradation. Both outcomes are scientifically valid results and shall be reported without suppression or sign correction.

5.5 This test method is structured to support incorporation by normative reference into external performance standards.

5.6 Unlike methods that rely on physical dimensions of image artifacts, this test method acknowledges that image quality is fundamentally defined by the ability of an observer to perform a clinical task.

5.7 The use of a fully specified and deterministic model observer (CHO) removes human-reader variability, providing a reproducible measure of how the MAR algorithm affects lesion detection performance.

6. Interferences

6.1 Variability in image processing parameters between MAR-enabled and MAR-disabled conditions will inflate or deflate ΔAUC and invalidate the test result.

6.2 Deviations from the specified Laguerre–Gauss channel templates, channel width parameter, Tikhonov regularization value, or covariance estimation procedure will reduce inter-laboratory reproducibility.

6.3 Alteration of the standardized dataset, including resampling, interpolation, windowing before CHO input, or truncation of the HU range, invalidates the test.

6.4 Use of floating-point arithmetic below 64-bit precision may introduce numerical bias exceeding the ±0.001 AUC equivalence tolerance.

6.5 Failure to verify the SHA-256 checksum prior to analysis invalidates the test result.

7. Apparatus

7.1 Imaging system or processing platform capable of applying MAR processing to the supplied volumetric dataset in DICOM CT format.

7.2 Computational platform capable of executing the reference 2D CHO analysis software.

7.3 Statistical software capable of Mann–Whitney AUC estimation and bootstrap confidence interval computation.

7.4 SHA-256 checksum verification utility.

7.5 The computational environment (software version, operating system, numerical libraries, and hardware platform) used for CHO analysis shall be documented in the test report.

8. Reagents and Materials

8.1 Standardized reconstructed digital volumetric dataset distributed with this standard, SHA-256 checksum verified per §11.1. The dataset consists of 160 DICOM CT series: 40 lesion-present / noMAR, 40 lesion-absent / noMAR, 40 lesion-present / MAR-ready, and 40 lesion-absent / MAR-ready, each comprising 256 axial slices at 512 × 512 × 0.5 mm isotropic.

8.2 Laguerre–Gauss channel template definitions as specified in §A1.4 and provided in machine-readable form with the dataset distribution.

8.3 Reference CHO implementation distributed with this standard. The reference implementation is the normative arbiter of correct CHO output. Alternative implementations shall demonstrate numerical equivalence within ±0.001 AUC on the supplied checksum-verified validation dataset before use.

8.4 Checksum manifest file (SHA-256) distributed with the dataset.

9. Hazards

9.1 No ionizing radiation exposure is required when using the supplied digital dataset.

10. Sampling, Test Specimens, and Test Units

10.1 The test unit consists of paired volumetric image datasets (MAR enabled and MAR disabled) derived from identical input data.

10.1.1 *Canonical Test Configuration*—This test method defines a single canonical configuration. All parameters are normative and fully specified in this section and in Annex A1. No additional configurations, parameter substitutions, or alternative geometries are permitted. The canonical configuration consists of:

Parameter	**Value**	**Notes**
Matrix (x, y, z)	512 × 512 × 256 voxels	Per A1.1
Voxel size	0.5 mm isotropic	FOV = 256 mm
Background HU	40 HU (soft tissue)	Uniform; noise added per realization
Metal rod diameter	10 mm	Full z-extent, centered at (256, 256)
Metal HU	3000 HU (fixed)	Not user-modifiable
Lesion disc diameter	5 mm	Slice 128 only; single representative slice
Lesion offset	5 mm beyond metal boundary, +x	Center at (281, 256)
Lesion contrast	~12 HU physics contrast	Established in sinogram-domain via μ_{lesion} no post-FBP hard-set
Background noise	30 HU Gaussian, IID	Applied only outside lesion and metal masks

Artifact model	Physics-based Poisson	60 keV monochromatic; see A1.7
Artifact jitter	Uniform (–15°, +15°) per realization	Independent per realization; see A1.7
Realizations	40 per condition	LP/LA × MAR/noMAR; 160 total

10.1.2 *Natural Reconstruction Rule*—The lesion signal shall be established exclusively in the sinogram domain via physics-based attenuation. No post-filtered backprojection (post-FBP) Hounsfield Unit (HU) replacement or "hard-set" overrides shall be applied to lesion voxels. The construction order within each realization slice shall be: (1) Forward Projection: Generate parallel-beam projections of the phantom background and, for lesion-present (LP) realizations, the lesion disc using the specified monochromatic attenuation coefficients (μ) at 60 keV; (2) Noise Modeling: Apply the *Vaishnav* noise model (Poisson photon statistics, scatter, and Gaussian electronic noise) to the raw projection data; (3) Reconstruction: Perform FBP reconstruction using a Ram-Lak filter and apply the DC calibration offset to ensure background tissue consistency; and (4) Metal Restoration: Restore voxels within the metallic object mask to 3000 HU as the final step, overriding all previously calculated values. The target effective lesion contrast of ~12 HU emerges as a natural consequence of the Radon inversion process. Violation of this construction order, or the application of any artificial pixel-replacement logic to the lesion ROI, produces a non-physical signal and invalidates the test result under this standard.

10.2 For the canonical test configuration, a minimum of 40 statistically independent lesion-present and 40 lesion-absent image volumetric realizations shall be analyzed per condition. The minimum total is therefore 160 volumetric image sets (40 LP-noMAR, 40 LA-noMAR, 40 LP-MAR, 40 LA-MAR).

10.3 Statistical independence shall be achieved using the predefined set of independent volumetric realizations provided with the standardized dataset, each generated with a unique, non-overlapping random seed. Laboratories shall not generate additional noise realizations or alter the provided image volumes. Each full 3D volume (256 slices) constitutes one independent realization. Individual slices within a volume, or multiple slices extracted from the same volume, are not independent samples and shall not be treated as such in CHO training or testing.

11. Preparation of Apparatus

11.1 Verify SHA-256 checksum of every file in the distributed dataset against the manifest provided with the standard. Any checksum mismatch disqualifies the dataset and invalidates any results derived from it.

11.2 Validate the CHO implementation against the supplied reference validation dataset, confirming AUC agreement within ±0.001.

11.3 Configure the imaging system MAR parameters as specified in the test plan. Document all configuration settings.

12. Calibration and Standardization

12.1 Validate CHO implementation using the supplied reference checksum-verified validation dataset prior to each test campaign. Validation shall be repeated whenever the CHO software or computational environment is updated.

12.2 No additional calibration of the imaging system is required or permitted under this standard.

13. Conditioning

13.1 No specimen conditioning is required. The dataset is digital and requires no physical preparation.

14. Procedure

14.1 Verify dataset checksums per §11.1.

14.2 Process the standardized noMAR volumetric dataset (20 LP and 20 LA volumes) through the imaging system with MAR disabled.

14.3 Process the identical standardized dataset through the imaging system with MAR enabled, using identical reconstruction and postprocessing parameters. The MAR enable/disable state is the only permitted difference.

14.4 Verify that the processed output volumes match the expected matrix dimensions, voxel size, and bit depth specified in §A1.1 before proceeding.

14.5 Apply the 2D CHO (§A1.5) to Slice 128 (LESION_SLICE_INDEX = 128, zero-indexed) only of the paired lesion-present and lesion-absent image volumes for each condition (MAR and noMAR), using the fixed ROI specified in §A1.5.4. Compute AUC values for each condition via the Mann-Whitney statistic. Three-dimensional volumetric integration across z is PROHIBITED under this standard.

14.6 Estimate covariance from lesion-absent volumes only, using the pooled sample covariance across all 40 LA realizations for the relevant condition.

14.7 Compute AUC for each condition via the Mann–Whitney statistic (§A1.5).

14.8 Calculate $\Delta AUC = AUC_{MAR} - AUC_{noMAR}$.

14.9 Compute bootstrap 95% confidence intervals per §A1.5(b).

14.10 Document all system settings, software versions, and any deviations from this procedure.

14.11 All preprocessing, reconstruction, and postprocessing parameters other than the MAR enable / disable state shall remain identical between conditions. Any parameter change, including window / level settings applied before CHO input, invalidates the test result.

15. Calculation or Interpretation of Results

15.1 $\Delta AUC = AUC_{MAR} - AUC_{noMAR}$

15.2 The reported test result shall be the mean ΔAUC across all replicates. ΔAUC shall be reported to three decimal places with sign.

15.3 Standard deviation of ΔAUC across replicates shall be reported.

15.4 Bootstrap 95% confidence intervals shall be reported.

15.5 The AUC estimation bias (difference between resubstitution and hold-out estimates) shall be reported per §A1.5(b).

15.6 Negative ΔAUC values shall be reported without modification. A negative result indicates that the MAR algorithm degraded lesion detectability relative to the no-MAR condition and is a scientifically valid and reportable outcome.

16. Report

16.1 The report shall include:

(a) System identification (manufacturer, model, software version)

(b) MAR algorithm name and version

(c) All image processing parameters applied during MAR-enabled and MAR-disabled conditions, with explicit confirmation that only the MAR state differed

(d) Dataset version identifier and SHA-256 manifest verification result

(e) CHO software version and validation AUC result (§12.1)

(f) Computational environment (OS, CPU/GPU, numerical library versions)

(g) Number of LP and LA replicates analyzed per condition

(h) Mean ΔAUC (three decimal places, with sign)

(i) Standard deviation of ΔAUC

(j) Bootstrap 95% confidence interval

(k) AUC estimation bias (resubstitution minus hold-out)

(l) Individual AUC_{MAR} and AUC_{noMAR} values

(m) Any deviations from this test method

17. Precision and Bias

17.1 *Precision*—The precision of this test method shall be determined by an interlaboratory study conducted in accordance with ASTM E691 and ISO 5725-2.

17.1.1 The following precision statistics shall be determined:

(a) Repeatability standard deviation (S_r)

(b) Reproducibility standard deviation (S_R)

(c) Repeatability limit ($r = 2.8 \times S_r$)

(d) Reproducibility limit ($R = 2.8 \times S_R$)

17.1.2 The precision statement shall be incorporated into this standard following completion of the full interlaboratory study.

17.1.3 The full interlaboratory study shall include a minimum of 6 laboratories in accordance with ASTM E691 recommendations, each processing the identical checksum-verified dataset.

17.1.4 The precision statement shall report S_r, S_R, r, and R values expressed in units of ΔAUC.

17.1.5 The digital, checksum-verified nature of this test method minimizes the resource barriers traditionally associated with interlaboratory studies by eliminating physical phantoms, shipping logistics, and clinical scanner time requirements.

17.1.6 Pilot precision data requirement: Before this standard proceeds to first ASTM ballot, the sponsoring subcommittee shall provide pilot precision data from a minimum of 3 laboratories demonstrating that: (a) mean ΔAUC is statistically distinguishable from zero for a MAR algorithm known to improve detectability; (b) the within-laboratory standard deviation of ΔAUC does not exceed 0.05 AUC units; and (c) the AUC estimation bias (resubstitution minus hold-out) does not exceed 0.02 AUC units at N=40 realizations.

17.2 Bias—Bias in the absolute sense cannot be determined because no accepted reference value for true lesion detectability exists independently of the measurement method. Systematic effects identified during interlaboratory evaluation shall be reported. Sources of systematic effect include: CHO implementation differences, floating-point accumulation errors, and covariance regularization choices.

18. Keywords

18.1 **metal artifact; metal artifact reduction; MAR; channelized Hotelling observer; model observer; ROC curve; AUC; ΔAUC; lesion detectability; interlaboratory study; precision and bias; tomographic imaging; computed tomography; signal-known-exactly; Laguerre–Gauss channels; reproducibility**

ANNEX

(Normative)

A1. Dataset Geometry, Lesion Specification, Observer Definition, and Statistical Procedures

A1.1 Volume Geometry:

(a) Matrix: 512 × 512 × 256 voxels (x, y, z)

(b) Voxel size: 0.5 mm isotropic in all three dimensions

(c) Field of view: 256 mm (512 voxels × 0.5 mm / voxel)

(d) HU range encoded as signed 16-bit integer (INT16) in DICOM pixel data

(e) Rescale slope: 1.0; Rescale intercept: 0.0 (HU = stored value)

A1.2 Phantom Cross-Section Geometry:

The phantom cross-section represents a simplified torso geometry consisting of an elliptical body region, a cylindrical metal rod, and (when present) a lesion disc. The metal rod is a uniform cylinder spanning the full z-extent of the volume to ensure consistent artifact generation. The lesion is a single disc confined to Slice 128 only.

(a) Body ellipse semi-axis x: 85 mm (170 voxels at 0.5 mm/voxel)

(b) Body ellipse semi-axis y: 60 mm (120 voxels at 0.5 mm/voxel)

(c) Body center: (x, y) = (256, 256) voxels (image center)

(d) Body interior HU: 40 HU (soft tissue equivalent)

(e) Exterior (outside body ellipse) HU: −1000 HU (air)

(f) Gaussian noise ($\sigma = 30$ HU) applied to body interior only, excluding lesion and metal masks, independently per realization per §A1.7

A1.3 Metal Object Specification:

(a) Geometry: right circular cylinder, full z-extent

(b) Diameter: 10 mm (radius = 5 mm = 10 voxels at 0.5 mm/voxel)

(c) Center: $(x_m, y_m) = (256, 256)$ voxels

(d) Fixed HU value: 3000 HU

(e) Metal boundary x-coordinate (positive x face): $x_m + r = 256 + 10 = 266$ voxels

(f) Metal voxels shall be restored to 3000 HU as the final step in slice construction, overriding noise and artifact

A1.4 Lesion Specification:

(a) Geometry: right circular disc, slice 128 only

(b) Diameter: 5 mm (radius = 2.5 mm = 5 voxels at 0.5 mm/voxel)

(c) Lesion center x-coordinate: $x_0 = x_m + r_m + 5\text{ mm} + r_{lesion} = 256 + 10 + 10 + 5 = 281$ voxels

(d) Lesion center y-coordinate: $y_0 = 256$ voxels

(e) Fixed HU value (SKE): ~12 HU effective FBP contrast (via MU_LESION_CM)

(f) Implementation: Lesion contrast is established in the sinogram domain only. No post-FBP HU replacement shall be applied to voxels within the lesion mask.

(g) Signal contrast: ~12 HU effective FBP contrast above soft tissue background

(h) Pre-observer per-voxel CNR: ~0.4 (noise-limited task)

A1.5 Channelized Hotelling Observer (CHO) Specification:

A1.5.1 Channel Type and Parameters

(a) Channel type: Laguerre–Gauss (LG) radial channels

(b) Number of channels: 10

(c) Channel order n: 0 through 9 (one channel per order)

(d) Channel width parameter a: 1.5 × r_{lesion} = 1.5 × 5 voxels = 7.5 voxels (3.75 mm). This value is fixed and not user-modifiable.

(e) Channel normalization: each channel $u_n(r)$ is L_2-normalized over the two-dimensional ROI domain such that $\|u_n\|_2 = 1$.

(f) The n-th 2D Laguerre–Gauss channel is defined as: $u_n(r) = L_n\left({2\pi r^2}/{a^2}\right) \times exp\left({-\pi r^2}/{a^2}\right)$, where L_n is the n-th Laguerre polynomial and *r* is the radial distance from the lesion center in voxels. The channel templates are strictly two-dimensional and shall be applied to Slice 128 only. Three-dimensional extension is prohibited.

A1.5.2 Covariance Estimation

(a) The CHO covariance matrix *K* shall be estimated from lesion-absent (LA) volumes exclusively.

(b) Covariance shall be estimated by pooling channel output vectors across all 40 LA realizations of the relevant condition (MAR or noMAR). Separate covariance matrices shall be estimated for the MAR and noMAR conditions.

(c) Regularization: Tikhonov regularization shall be applied as $K_{reg} = K + \lambda I$, where $\lambda = 0.01 \times \text{trace}(K) / p$, p = number of channels = 10. This normalization ensures λ scales with the data and is invariant to HU units. λ is fixed by this formula and shall not be user-modified.

(d) Internal Observer Noise: To match human observer thresholds, an internal noise variance shall be added to the diagonal of the covariance matrix: $K_{total} = K_{ext} + \sigma_{int}^2 I$, where $\sigma_{int} = 15$. Omission of this parameter invalidates interlaboratory comparability.

A1.5.3 Observer Dimensionality:

The CHO shall operate on a two-dimensional (2D) region of interest from Slice 128 only. Three-dimensional volumetric integration across z is PROHIBITED.

A1.5.4 Region of Interest (ROI)

(a) ROI x-extent: lesion center x ± 60 voxels = 221 to 341 (121 voxels)

(b) ROI y-extent: lesion center y ± 60 voxels = 196 to 316 (121 voxels)

(c) ROI z-extent: all 256 slices (only slice 128 of volume)

(d) ROI center coordinates are fixed per this annex and shall not be adjusted between conditions or

realizations.

(e) The same ROI shall be used for MAR-enabled and MAR-disabled conditions.

A1.6 Statistical Procedures

(a) AUC shall be estimated via the Mann–Whitney statistic applied to CHO test statistics from LP and LA volumes.

(b) Ties in the Mann–Whitney statistic shall be handled using mid-rank assignment.

(c) Bootstrap confidence intervals shall be computed using 1000 resamples (sampling LP and LA test statistics jointly with replacement). The 2.5th and 97.5th percentiles of the bootstrap AUC distribution define the 95% CI.

(d) AUC estimation bias shall be quantified using resubstitution and hold-out CHO training / testing strategies per Wunderlich and Noo (*IEEE Trans Med Imaging*, 34(2), 2015). The bias estimate is defined as $b = AUC_{resubstitution} - AUC_{hold\text{-}out}$. Both estimates and the bias shall be reported.

(e) Minimum 64-bit (double-precision) floating-point arithmetic shall be used throughout CHO computation, including channel projection, covariance estimation, matrix inversion, and test statistic computation.

A1.7 Background Variability Mechanism

A1.7.1 The canonical dataset provides background variability via two independent mechanisms: (i) per-realization artifact jitter, and (ii) independent Gaussian noise realizations.

A1.7.2 *Artifact jitter*—For each realization i (i = 1 to 40), a jitter angle θ_i shall be drawn from a uniform distribution on [−15°, +15°]. The base artifact template (§A1.7.4) shall be rotated by θ_i about the image center using bilinear interpolation. The rotated template shall be clipped to the body mask and zeroed within the metal mask before addition.

A1.7.3 *Noise realizations*—Independent Gaussian noise ($\sigma = 30$ HU, $\mu = 0$) is added to each realization using a unique random seed. For realization i, the seed shall be BASE_SEED + i where BASE_SEED is a fixed integer specified in the dataset metadata. Noise is applied only within the body mask and outside the lesion and metal masks.

A1.7.4 *Artifact template*—The base artifact template is a 2D HU field computed once by: (1) forward projecting a phantom containing the body background and metal rod using 180 equally-spaced projection angles over [0°, 180°]; (2) simulating photon starvation by replacing sinogram values above the 99th percentile of non-zero values with 2% of the 50th percentile; (3) reconstructing both the original and corrupted sinograms using FBP with a Ram-Lak filter; (4) computing the difference (corrupted minus original); (5) zeroing the template within the metal and outside the body mask; (6) scaling so that the maximum absolute value within the body (excluding metal) is 400 HU.

A1.7.5 The artifact template, jitter angles, and noise seeds for all 40 realizations are fixed in the reference dataset. Laboratories shall use the provided dataset and shall not regenerate these values.

A1.8 Reproducibility Requirements

The following parameters shall not be modified under any circumstance. Modification of any parameter invalidates the test result under this standard:

Parameter	Specified Value	Location
Voxel size	0.5 mm isotropic	§A1.1
Matrix dimensions	512 × 512 × 256	§A1.1
LG channel type	Laguerre–Gauss, n = 0..9	§A1.5.1(a-c)
Channel width parameter a	7.5 voxels (1.5 x r_{lesion})	§A1.5.1(d)

Channel normalization	L2-normalized	§A1.5.1(e)
Internal noise σ	15 (Normative)	§A1.5.2(d)
Tikhonov λ	0.01 x trace(K)/p	§A1.5.2(c)
Covariance source	LA volumes only, pooled	§A1.5.2(a,b)
Observer dimensionality	2D, Slice 128 only	§A1.5.3
ROI dimensions	121 × 121 voxels (2D)	§A1.5.4
ROI center	(281, 256) voxels in xy	§A1.4(c,d)
Lesion contrast	~12 HU (Sinogram-domain)	§A1.4(e,f)
Lesion geometry	Single disc, Slice 128 only	§A1.4(a)
Metal HU	3000 HU (restored last)	§A1.3(d,f)
Artifact peak HU	400 HU	§A1.7.4
Artifact jitter range	Uniform [–15°, +15°]	§A1.7.2
Realizations per condition	40 minimum (160 total)	§10.2
Floating-point precision	64-bit minimum (double)	§A1.6(e)
Baseline AUC_{noMAR}	0.7063 (Validated 03/14/2026)	§1A.2
SHA-256 checksum	Must match manifest	§8.4, 11.1
CHO numerical equivalence	±0.001 AUC on validation set	§8.3

449

450 The result shall not be reported as compliant with this standard if any of the above parameters
451 have been modified.